

WASP-52b

R-0583

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_181206 · created 2026-07-24T19:01:53+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458458.5954 +/- 0.0039 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.216 +/- 0.055
Transit depth (Rp/Rs)^2	4.64 +/- 2.38 [%]
Orbital Inclination (inc)	83.09 +/- 1.81
Airmass coefficient 1 (a1)	0.989 +/- 0.019
Airmass coefficient 2 (a2)	0.011 +/- 0.073
Scatter in the residuals of the lightcurve fit is	1.89 %
Best Comparison Star	#3 - [242, 358]
Optimal Aperture	3.69
Optimal Annulus	4.38
Transit Duration (day)	0.045 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.216 ± 0.055 vs 0.1646 (deviation 0.93σ)
Duration fit vs published	0.045 d vs 0.0758 d (deviation 1.71σ)
Mid-time O–C	-4.3 min
Depth SNR	3.9
Residual scatter	1.89 %

Light curve

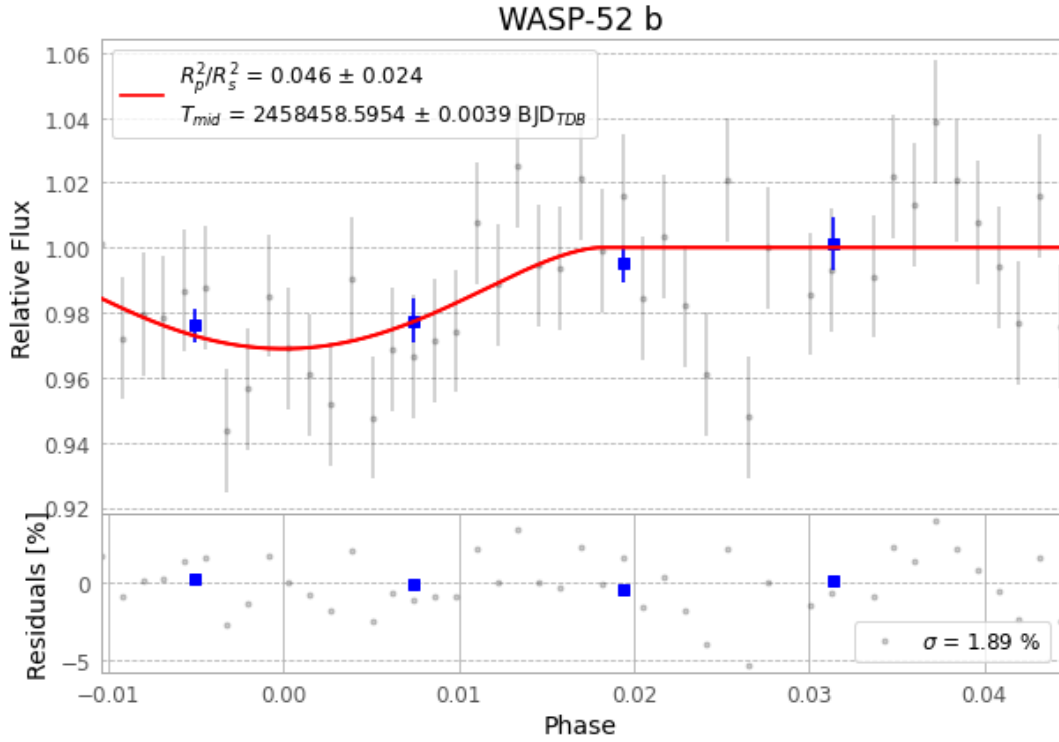


Figure 1 — FinalLightCurve_WASP-52 b_2018-12-05.png (SHA-256 3fd510cb77be2944...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [266, 287], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 53aac2b9ce02fec8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

49 FITS frames (32 MB), WASP-52181206014812.FITS → WASP-52181206041213.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-181206.log (e5ea288cc9ce...), FinalLightCurve_WASP-52 b_2018-12-05.png (3fd510cb77be...), FinalLightCurve_WASP-52 b_2018-12-05.csv (bbf229c77fbd...)

Compute resources

Wall time 255.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-24 19:01Z.